Well-quasi-ordered classes of bounded clique-width

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Abstract

We study classes of graphs with bounded clique-width that are well-quasi-ordered by the induced subgraph relation, in the presence of labels on the vertices. We prove that, given a finite presentation of a class of graphs, one can decide whether the class is labelled-well-quasi-ordered. This answers positively to two conjectures of Pouzet in the restricted case of bounded clique-width classes. Namely, we prove that being labelled-well-quasi-ordered by a set of size 2 or by a well-quasi-ordered infinite set are equivalent conditions, and that in such cases, one can freely assume that the graphs are equipped with a total ordering on their vertices. Finally, we provide a structural characterization of those classes as those that are of bounded clique-width and do not existentially transduce the class of all finite paths.

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This document uses knowledge: a notion points to its *definition*.

1 Introduction

The theory of *well-quasi-orderings* (WQOs) provides a powerful combinatorial setting that has found applications in various areas of mathematics and computer science. In graph theory, the celebrated result of Robertson and Seymour [26] states that the class of all finite graphs is well-quasi-ordered under the minor relation, a profound result with deep algorithmic consequences. WQOs are also at the heart of *well-structured transition systems*, infinite-state transition systems over which verification algorithms can be designed [2, 3]. One appealing feature of WQOs is that they are closed under various operations, such as the sum and the product of WQOs. As an example, Higman's lemma [20] states that if X is a WQO, then the set X^* of finite words over X is also a WQO under the subword embedding relation, and this result has been used in the verification of so-called *lossy channel systems* [1].

Undirected finite graphs are naturally equipped with the *induced subgraph relation*, where G is an induced subgraph of H if G can be obtained from H by deleting vertices. Unlike the graph minor relation, the induced subgraph relation is not a well-quasi-ordering on the class of all finite graphs, as witnessed by the infinite family of cycles of increasing size, which form an infinite sequence of pairwise incomparable graphs. However, some classes of finite graphs are well-quasi-ordered under the induced subgraph relation: for instance, the class of all finite paths, or the class of all finite cliques. Distinguishing which classes of (finite) graphs are well-quasi-ordered under the induced subgraph relation is a long-standing open problem in graph theory.

In general, there is little hope of characterising all such classes, since any countable order with finite infixes can be embedded into the induced subgraph relation on finite graphs (this

Class	WQO	2-WQO	$\forall$-WQO
Cycles	×	×	×
Paths	✓	×	×
Cliques	✓	✓	✓

Figure 1 Well-quasi-ordering properties of classes in Example 1. No example 2-WQO but not $\forall$-WQO is known.

is a consequence of [22, Lemma 5.1]). If one considers *labelled* well-quasi-orderings on classes of finite graphs, then there are high hopes of being able to provide a full characterisation of (labelled) well-quasi-ordered classes. Given a class $\mathcal{C}$ of finite graphs, and a well-quasi-ordering $(X, \leq)$, one can consider the class $\mathsf{Label}_X(\mathcal{C})$ of graphs in $\mathcal{C}$ whose vertices are labelled by elements of X, that is, equipped with a function $\ell \colon V(G) \to X$. The induced subgraph relation is then extended to labelled graphs by requiring that the labels are preserved by the embedding, i.e. that the label of a vertex in the smaller graph is less than or equal to the label of its image in the larger graph. A class $\mathcal{C}$ of finite graphs is said to be *$(X, \leq)$-well-quasi-ordered* if the class $\mathsf{Label}_X(\mathcal{C})$ is well-quasi-ordered under the labelled induced subgraph relation. We will say that it is *k-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered where X is a k-element antichain, and that it is *$\forall$-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered for every well-quasi-ordering $(X, \leq)$.

▶ **Example 1.** The class Cycles of all finite cycles is not well-quasi-ordered, since the infinite sequence of cycles forms an infinite antichain. The class Paths of all finite paths is well-quasi-ordered since $(\mathsf{Paths}, \subseteq_i)$ is isomorphic to $(\mathbb{N}, \leq)$. However, it is not 2-well-quasi-ordered, since the infinite sequence of paths with coloured endpoints forms an infinite antichain. The class Cliques of all finite cliques is $\forall$-well-quasi-ordered, since any X-labelled clique can be identified with a finite multiset of labels from X, and the set of finite multisets over a well-quasi-order is itself well-quasi-ordered by standard results on well-quasi-orders [12].

The study of labelled structures is a recurring theme in the theory of well-quasi-orderings: classical results such as Higman's lemma [20] or Kruskal's tree theorem [21] are actually stating that some classes of finite relational structures (respectively finite linear orders and finite meet trees) are well-quasi-ordered under the labelled embedding relation, for every well-quasi-ordering of labels. It was conjectured by Pouzet [24] that being $\forall$-well-quasi-ordered reduces to being 2-well-quasi-ordered for hereditary classes of structures (see [25, Problems 9, (1)] for a modern formulation). Another interesting conjecture of Pouzet [25, Problems 12] states that when a class $\mathcal{C}$ is $\forall$-well-quasi-ordered, one can add a total order on the vertices of its structures such that the new class is still $\forall$-well-quasi-ordered. This conjecture gained interest in the recent development of algebraic methods for polynomials over relational structures [19, Section 3, Theorem 11 and 12], where "monomials" are defined as structures labelled by $(\mathbb{N}, \leq)$. We refer to this property as being *orderably $\forall$-well-quasi-ordered*.

Even for classes of finite undirected graphs, these conjectures remain open. Several works dating back to the 90s have identified structural properties of hereditary classes of finite graphs that guarantee that they are $\forall$-well-quasi-ordered, such as being of bounded tree-depth [15]. In 2010, Daligault, Rao and Thomassé [11] initiated the study in the reverse direction, by considering a syntax for graph classes (using NLC graph expressions), and identifying which ones among them are 2-well-quasi-ordered. In this context, they verified Pouzet's conjectures, but were unable to generalise their results to arbitrary graph subclasses [11, Conjecture 4]. All the classes considered in [11] have what is called *bounded clique-width*,[1] and it is believed that it is enough to understand hereditary classes of bounded clique-width to solve these conjectures in full generality, as stated in the following conjecture.

▶ **Conjecture 2** ([11, Conjecture 5]). *Let $\mathcal{C}$ be a hereditary class of finite graphs that is 2-well-quasi-ordered. Then, $\mathcal{C}$ is of bounded clique-width.*

In parallel with Pouzet's conjectures, another line of research has focused on finding structural obstructions to being ∀-well-quasi-ordered. The natural candidate for such an obstruction is the class of all finite paths (see Figure 1). For the particular classes studied in [11], it was shown that every class that is not 2-well-quasi-ordered contains all finite paths [11, Corollary 2]. There are examples of hereditary classes that are not 2-well-quasi-ordered and do not contain all finite paths, but in every known such example, one can "extract" finite paths using very simple logical formulae. This led to the following conjecture, where existentially transduces is a notion from model theory (see Section 2 for a formal definition).

▶ **Conjecture 3** ([23, Conjecture 26]). *Let $\mathcal{C}$ be a hereditary class of finite graphs. If $\mathcal{C}$ is not 2-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

In 2024, Lopez [23] studied labelled well-quasi-orderings on classes of graphs of bounded *linear* clique-width, leveraging combinatorial tools from automata theory (Simon's factorisation forests [27]). This yielded a weakening of Pouzet's conjectures [23, Corollary 2], namely, that a hereditary class of bounded linear clique-width is ∀-well-quasi-ordered if and only if it is k-well-quasi-ordered for every $k \in \mathbb{N}$. It is apparent from the proof that the techniques also yield a natural ordering on the vertices whenever the class is ∀-well-quasi-ordered, although this is not formally stated in the paper. It was conjectured [23, Section 5] that these techniques could be refined to work on classes of bounded clique-width (not necessarily linear), and actually obtain the correspondence between 2-well-quasi-ordering and ∀-well-quasi-ordering.

Contributions

Our main contribution is Theorem 4, which answers positively to the conjectures of [23]. To that end, we first consider a generalisation of the framework of [11] from NLC graph expressions to MSO-interpretations from trees (these will be formally defined in Section 2). This also generalises the setting of [23], which studied MSO-interpretations from *words*. Our main combinatorial result is the following.

▶ **Theorem 4.** *Let $\mathcal{C}$ be the image of a class of finite trees under an* MSO *interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

1. *$\mathcal{C}$ is 2-well-quasi-ordered,*
2. *$\mathcal{C}$ is ∀-well-quasi-ordered,*
3. *$\mathcal{C}$ is orderably ∀-well-quasi-ordered,*

From Theorem 4, we derive the following theorem that solves Pouzet's conjectures for hereditary classes of bounded clique-width. This trades the decidability of Theorem 4 to handle more classes.[2]

[1] This will be defined formally in Section 2

[2] There are uncountably many hereditary classes of bounded clique-width, but only countably many images of trees under MSO interpretations.

▶ **Theorem 5.** *The conclusion of Theorem 4 holds when $\mathcal{C}$ is a hereditary class of graphs having bounded clique-width.*

Because our proofs are constructive and combinatorial, we can derive from them obstructions to being $\forall$-well-quasi-ordered, which we list in Theorem 6. In our statements, we use infinite antichains with a periodic structure that were called *periodic antichains* in [4, Section 7]. These antichains appear in [4, Conjecture 2], that is concerned with classes of graphs defined by finitely many excluded induced subgraphs, and relates the presence of these antichains to the fact that they are well-quasi-ordered. We show that in our setting, containing such periodic antichains is equivalent to being not 2-well-quasi-ordered. The precise definitions of regular antichains and periodic antichains will be given in Section 6.

▶ **Theorem 6.** *Let $\mathcal{C}$ be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

1. *$\mathcal{C}$ is not 2-well-quasi-ordered,*
2. *There exists a class $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width which is not 2-well-quasi-ordered,*
3. *There exists a regular antichain in $\mathcal{C}$,*
4. *There exists a periodic antichain in $\mathcal{C}$,*
5. *$\mathcal{C}$ existentially transduces the class of all finite paths.*

We strongly believe that the results of Theorem 5 extend to non-hereditary classes of bounded clique-width graphs, but for the sake of clarity and space, we only state and prove them on hereditary ones. On the other hand, let us remark that Theorem 6 does not immediately extend to non-hereditary classes, as witnessed by the following example.

▶ **Example 7.** The class $\mathcal{C}$ of complete binary trees has bounded clique-width, is not 2-well-quasi-ordered, but contains no periodic antichain, and every subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width is 2-well-quasi-ordered. Furthermore, $\mathcal{C}$ existentially transduces the class of all finite paths.

Proof Sketch. It is well known that the class of complete binary trees has bounded clique-width. Furthermore, it is not 2-well-quasi-ordered, since one can colour the leaves and the root of the complete binary trees to form an infinite antichain. On the other hand, any subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width must be finite: indeed, complete binary trees have unbounded linear clique-width. Finally, because periodic antichains have bounded linear clique-width (see Section 6) there cannot be any periodic antichain in $\mathcal{C}$. The existential transduction of all finite paths from $\mathcal{C}$ is obtained by existentially transducing all induced subgraphs (see Example 9), which in particular contains all finite paths. ◀

Outline

Section 2 introduces the necessary background on well-quasi-orders, graphs, MSO interpretations and first-order transductions. Section 3 shows how to reduce the study of graph classes obtained by MSO interpretations from trees to the study of trees labelled over finite monoids. This section also shows how to leverage forward Ramseyan splits to obtain well-behaved tree representations of graphs that are well-quasi-ordered (as trees) with respect to the so-called *gap embedding* relation. Section 4 identifies combinatorial obstructions to being 2-well-quasi-ordered in images of interpretations from trees, and is the core combinatorial part of the paper. Section 5 brings together the previous results to prove our Theorem 4 and Theorem 5. We also discuss the equivalence between the first two items of Theorem 6. The remaining items of Theorem 6 are proven equivalent in Section 6. Finally, in Section 7 we discuss some perspectives and open problems.

2 Preliminaries

We assume the reader to be familiar with basic notions of logic on finite relational structures, such as monadic second-order logic (MSO), first-order logic and quantifier-free formulae.

Graphs A graph G consists of a finite set $V(G)$ of vertices and a set $E(G) \subseteq V(G)^2$ of edges. A graph is *undirected* if $(u,v) \in E$ implies $(v,u) \in E$ for all $u,v \in V$, and directed otherwise. In this paper we will focus on finite undirected graphs without self-loops. Given a set X, an *X-labelled graph* is a graph equipped with a function $\ell\colon V(G) \to X$ that assigns a label to each vertex. Given a class $\mathcal{C}$ of finite undirected graphs, write $\mathsf{Label}_X(\mathcal{C})$ for the class of freely X-labelled graphs in $\mathcal{C}$, i.e., the class of all X-labelled graphs (G,ℓ) such that $G \in \mathcal{C}$.

Whenever $(X,\leq)$ is a quasi-order, we say that a graph G *embeds* into a graph H if there exists a *monomorphism* from G to H, i.e., an injective function $f\colon V(G) \to V(H)$ such that for all $u,v \in V(G)$, $(u,v) \in E(G)$ if and only if $(f(u),f(v)) \in E(H)$, and such that for all $v \in V(G)$, we have $\ell_G(v) \leq \ell_H(f(v))$. We write $G \subseteq_i H$ to denote that G embeds into H, which we also call the *induced subgraph relation*. When no order is specified on a set X, the order relation is assumed to be equality, i.e. $(X,=)$.

A class of graphs $\mathcal{C}$ is a *hereditary class* if it is closed under taking induced subgraphs, i.e. for $G \in \mathcal{C}$, if $H \subseteq_i G$, then $H \in \mathcal{C}$. The *hereditary closure* of a class of graphs $\mathcal{C}$ is the smallest hereditary class containing $\mathcal{C}$.

An *ordered graph* is a finite undirected graph G equipped with a linear order $\leq_G$ on its vertices. The notion of induced subgraph relation extends to ordered graphs by requiring that the embedding $f\colon V(G) \to V(H)$ preserves the order, i.e., for all $u,v \in V(G)$, $u \leq_G v$ if and only if $f(u) \leq_H f(v)$.

Well-quasi-orderings A quasi-ordered set $(X,\leq)$ is a set X with a reflexive and transitive relation $\leq$. A sequence $(x_i)_{n\in\mathbb{N}}$ of elements in X is *good* if there exist $i < j$ such that $x_i \leq x_j$. A quasi-ordered set is *well-quasi-ordered* (*wqo*) if every infinite sequence is good. A *bad sequence* is an infinite sequence that is not good. A sequence of pairwise incomparable elements is called an *antichain*.

As mentioned in the introduction, several notions of well-quasi-ordering can be defined for a given class of finite undirected graphs $\mathcal{C}$. Let us briefly state them: $\mathcal{C}$ is *well-quasi-ordered* if the class $(\mathcal{C},\subseteq_i)$ is well-quasi-ordered, $\mathcal{C}$ is *k-well-quasi-ordered* for some $k \in \mathbb{N}$, if for every set $(X,=)$ of size k, the class $(\mathsf{Label}_X(\mathcal{C}),\subseteq_i)$ is well-quasi-ordered, and $\mathcal{C}$ is *$\forall$-well-quasi-ordered* if for every well-quasi-order $(X,\leq)$, the class $(\mathsf{Label}_X(\mathcal{C}),\subseteq_i)$ is well-quasi-ordered.

Finally, a class of graphs is *orderably well-quasi-ordered* (resp. *orderably k-well-quasi-ordered*, resp. *orderably $\forall$-well-quasi-ordered*) if there is a way to equip each graph G in $\mathcal{C}$ with a linear order $\leq_G$ on its vertices such that the resulting class of ordered graphs is well-quasi-ordered (resp. k-well-quasi-ordered, resp. $\forall$-well-quasi-ordered).

Trees In this paper, *trees* are finite, binary, and ordered (they have *left* and *right children*). They are understood as finite relational structures over the signature $\sigma = (\sqsubseteq, \preceq)$ where $\sqsubseteq$ is the *ancestor relation*, and $\preceq$ is the *sibling order* on the nodes of the tree. In a tree $\mathcal{T}$, the *root* is denoted by $\mathsf{root}(\mathcal{T})$, with $\mathcal{T}$ being left implicit when clear from context. The set of *leaves* is denoted by $\mathsf{Leaves}(\mathcal{T})$. Given two nodes x and y in a tree T, their *least common ancestor* is denoted by $\mathsf{lca}(x,y)$. We will often use the *parent relation*, which states that y is the unique immediate ancestor of x.

As in the case of graphs, we will consider vertex-labelled trees where labels are taken from a well-quasi-order $(X, \leq)$. We will also consider edge-labelled trees, typically using a finite set of labels Σ. The class of vertex-labelled and edge-labelled trees is denoted by Trees_Σ^X. The sets Σ or X are omitted when unused (i.e., Trees denotes the class of unlabelled trees). Given two nodes x, y such that x is an ancestor of y, the notation $[x : y]_{\mathcal{T}}$ denotes the *word* in Σ^* obtained by reading the labels of the edges on the path from x to y in $\mathcal{T}$; we leave $\mathcal{T}$ implicit when clear from context.

Bounded Clique-Width An MSO-*interpretation* $\mathcal{I}$ from Trees_Σ to finite undirected graphs is given by a tuple of MSO formulae that specify how to, respectively, select a subset of the leaves of the tree to be the vertices of the graph ($\phi_{\mathsf{univ}}(x)$), define the edges of the graph ($\phi_{\mathsf{edge}}(x, y)$), and define the domain of the interpretation (ϕ_{dom}). An interpretation $\mathcal{I}$ is called *simple* if $\phi_{\mathsf{univ}}(x)$ and ϕ_{dom} are tautologies.

▶ **Example 8.** An MSO-interpretation $\mathcal{I}$ from Trees to finite undirected graphs that produces all paths can be defined as follows: ϕ_{dom} is the formula that holds on linear trees (i.e., trees where every node has at most one non-leaf child), $\phi_{\mathsf{univ}}(x)$ is the formula that is true if and only if x is a leaf and its sibling is not a leaf, $\phi_{\mathsf{edge}}(x, y)$ is the formula that is true if and only if x and y are at distance exactly 3.

A class of graphs has *bounded clique-width* if it is included in the image of some MSO-interpretation from finite trees to finite undirected graphs [10]. It has *bounded linear clique-width* if it is included in the image of some MSO-interpretation from linear trees to finite undirected graphs. Equivalently, a class of graphs has bounded linear clique-width if it is included in the image of some simple MSO-interpretation from finite words to finite undirected graphs.

Transductions from graphs to graphs Another notion of logical transformation relevant in this paper is that of existential transduction and existential interpretation that were mentioned in Theorem 6 and Conjecture 3, and are defined in a similar way as MSO-interpretations.

An *existential interpretation* from (labelled) graphs to graphs is given by an existential first-order formula $\phi_{\mathrm{univ}}(x)$ that selects the vertices of the interpreted graph, an existential first-order formula $\phi_{\mathrm{edge}}(x, y)$ that defines the edges of the interpreted graph, and an existential first-order formula ϕ_{dom} that selects the graphs of $\mathcal{C}$ on which the interpretation is defined. The semantics is that for every graph $G \in \mathcal{C}$ satisfying ϕ_{dom}, the interpreted graph $\mathcal{I}(G)$ has as vertex set the vertices of G satisfying ϕ_{univ}, and has an edge between two vertices (u, v) if and only if G satisfies $\phi_{\mathrm{edge}}(u, v)$ (which is assumed to be symmetric and irreflexive on the selected vertices). A class $\mathcal{C}$ *existentially interprets* a class $\mathcal{D}$ if there exists an existential interpretation $\mathcal{I}$ from $\mathcal{C}$ to graphs such that $\mathcal{D} \subseteq \mathcal{I}(\mathcal{C})$. We say that $\mathcal{C}$ *existentially transduces* $\mathcal{D}$ if there exists a finite labelling set Σ such that $\mathsf{Label}_\Sigma(\mathcal{C})$ existentially interprets $\mathcal{D}$.

▶ **Example 9.** Let $\mathcal{C}$ be a class of graphs. Then, $\mathcal{C}$ existentially transduces the hereditary closure of $\mathcal{C}$, using labels $\Sigma = \{\circ, \bullet\}$, and the interpretation defined as follows: ϕ_{dom} is the tautological formula, $\phi_{\mathrm{univ}}(x)$ is the formula that is true if and only if x is labelled with $\bullet$, $\phi_{\mathrm{edge}}(x, y)$ is the formula that is true if and only if there is an edge between x and y in the original graph.

Let us remark that if a (non-hereditary) class of graphs having bounded clique-width is 3-well-quasi-ordered, then its hereditary closure is 2-well-quasi-ordered, and therefore the class itself is $\forall$-well-quasi-ordered. In particular, one gets a generalisation of Theorem 4 (resp.

Theorem 5) to non-hereditary classes of graphs of bounded clique-width, up to replacing 2-well-quasi-ordered by 3-well-quasi-ordered. As stated in the introduction, we strongly believe that our results can be extended to non-hereditary classes of graphs of bounded clique-width without this increase in the number of labels.

3 From Interpretations to Nested Trees

As a first step towards proving Theorem 4, we transform a simple MSO-interpretation from trees to graphs into a more combinatorial object that renders the interpretation *quantifier-free*. To that end, we chain two standard transformations from automata theory: we first convert an MSO-interpretation into a so-called monoid interpretation, and then factorise the trees using forward Ramseyan splits (nested trees) [7]. Finally, we show that the natural ordering on nested trees (the gap embedding) is *almost* order-preserving for the resulting interpretation from nested trees to graphs.

Let us fix for the rest of the section a finite alphabet Σ and a simple MSO-interpretation $\mathcal{I}$ from Trees_Σ to finite undirected graphs defined by a formula $\varphi(x, y)$ that defines the edges of the graphs.

3.1 From Interpretation to Monoids

It follows from standard arguments that to an MSO formula $\varphi(x, y)$ over Trees_Σ one can associate a finite monoid M and a morphism $\mu\colon \Sigma^* \to M$ such that for any tree T and any pair of leaves $x \preceq y$ in T, whether $T \models \varphi(x, y)$ is entirely determined by the values of $\mu([z : x]_\mathcal{T})$, $\mu([z : y]_\mathcal{T})$ and $\mu([\mathsf{root} : z]_\mathcal{T})$ where $z = \mathsf{lca}(x, y)$ is the least common ancestor of x and y. We refer to the book on Tree Automata Techniques and Applications for a comprehensive overview of the connection between Monadic Second Order Logic on trees, and tree automata [9]. In the rest of the section, we fix such a monoid M and morphism μ. To simplify notations, we write $[\![x : y]\!]_\mathcal{T}$ for $\mu([x : y]_\mathcal{T})$, that is, the product of the labels of the edges on the path from x to y in $\mathcal{T}$. If the path is empty (i.e., $x = y$), then $[\![x : y]\!]_\mathcal{T}$ is the identity element of M.

As a consequence, one can collect in a subset $P \subseteq M^3$ all the triples (a, b, c) that correspond to satisfying assignments of the formula $\varphi(x, y)$, as illustrated in Figure 2. We define a *monoid interpretation* from trees labelled over M to finite undirected graphs to be a tuple (Σ, μ, M, P) where Σ is a finite alphabet, $\mu\colon \Sigma^* \to M$ is a morphism to a finite monoid M, and $P \subseteq M^3$ is a set of triples of elements of M. The interpretation works as follows: given a tree $\mathcal{T}$ whose edges are labelled by elements of Σ, the vertices of the interpreted graph are the leaves of T, and there is an edge between two leaves $x \preceq y$ if and only if the triple $([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : y]\!]_\mathcal{T})$ belongs to P.

Adding the function $[\![x : y]\!]_T$ to the signature of trees, and assuming that lca is also part of the signature, we can rewrite the formula $\varphi(x, y)$ as a quantifier-free formula as follows:

$$([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : y]\!]_\mathcal{T}) \in P \tag{1}$$

The main idea driving this paper is that deciding when a class of graphs is $\forall$-well-quasi-ordered reduces to finding a suitable well-quasi-order on trees such that the interpretation $\mathcal{I}$ is *order-preserving* from trees to graphs, leveraging the standard Fact 10.

▶ **Fact 10** (Folklore). *Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two quasi-orders, and $f\colon X \to Y$ be a surjective order-preserving map. If $(X, \leq_X)$ is well-quasi-ordered, then $(Y, \leq_Y)$ is well-quasi-ordered.*

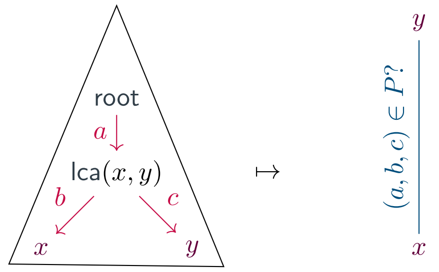

Figure 2 The interpretation of a tree using a monoid M and an accepting part $P \subseteq M^3$.

Similarly, if $f\colon X \to Y$ is an order reflection, that is, for every x_1, x_2 in X, $f(x_1) \leq_Y f(x_2)$ implies $x_1 \leq_X x_2$, and $(Y, \leq_Y)$ is well-quasi-ordered, then $(X, \leq_X)$ is well-quasi-ordered.

One candidate ordering on trees is the *composition ordering*. Given two trees $\mathcal{T}_1$ and $\mathcal{T}_2$ edge-labelled over a monoid M, we say that $\mathcal{T}_1$ is less than $\mathcal{T}_2$ in the composition ordering, written $\mathcal{T}_1 \preceq_{\mathsf{cmp}} \mathcal{T}_2$, if there exists a map $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that maps leaves to leaves and respects lca, $\preceq$, and $[\![\cdot : \cdot]\!]$. This ordering naturally extends to node-labelled trees by requiring that for every node x in $\mathcal{T}_1$, the label of x is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$. It is straightforward to check that the interpretation $\mathcal{I}$ is order-preserving from trees equipped with the composition ordering to graphs equipped with the induced subgraph ordering. Hence, to show that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered, it suffices to prove that the class of trees is well-quasi-ordered under the composition ordering when labelled using a well-quasi-ordered set $(X, \leq_X)$.

Historically, this approach has been used to prove that some classes of graphs are $\forall$-well-quasi-ordered; see for instance the results of [15, 11]. Unfortunately, the composition ordering on trees is often much stricter than the induced subgraph ordering. It was observed that the composition ordering on trees is well-quasi-ordered if and only if *for every* $P \subseteq M^3$, the class of graphs obtained from trees by considering only the triples in P is $\forall$-well-quasi-ordered [23, Theorem 24]. To understand what happens for a specific choice of P, we will develop a finer understanding of the trees themselves.

3.2 Forward Ramseyan Splits

The combinatorial ingredient that allows us to further decompose the trees comes from an adaptation of the classical Simon Factorisation Theorem for semigroups [27], adapted to trees by Colcombet [7]. The rest of this section explains how this factorisation works and how it can be used to define a more suitable ordering on trees.

A *split of height N* of a tree $\mathcal{T}$ is a mapping $\mathfrak{s}$ from the nodes of $\mathcal{T}$ to $\{1, \dots, N\}$. Given a split $\mathfrak{s}$ and two nodes $x \sqsubseteq_{\mathcal{T}} y$, we define $\mathfrak{s}(x\colon y)$ to be the minimal value of $\mathfrak{s}(z)$ for $x \sqsubset_{\mathcal{T}} z \sqsubset_{\mathcal{T}} y$, and $N+1$ otherwise. In particular, if $x = y$ or y is a child of x, then $\mathfrak{s}(x\colon y) = N+1$.

Let us fix a branch B of $\mathcal{T}$, that is, a maximal set of nodes of $\mathcal{T}$ that are pairwise comparable for $\sqsubseteq$. Two nodes x and y of B are *directed k-neighbours* if $\mathfrak{s}(x) = \mathfrak{s}(y) = k$ and $\mathfrak{s}(x\colon y) \geq k$. These two nodes are *$k$-neighbours* if they are *directed k-neighbours* in either direction. This defines an equivalence relation on the nodes of B for every $k \in \{1, \dots, N\}$, and a split thus induces a hierarchical structure on each branch based on its *k-neighbourhoods*. We illustrate the situation in Figure 3. We refer to the value of $\mathfrak{s}(x)$ as the *split depth* of the node x.

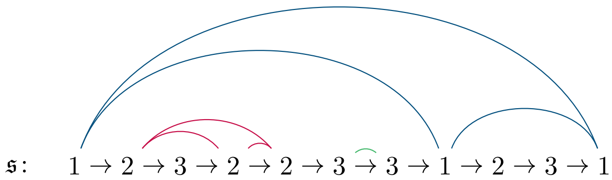

Figure 3 A split $\mathfrak{s}$ on a branch of a tree. The arcs in blue connect 1-neighbours, the arcs in red connect 2-neighbours, and the arcs in green connect 3-neighbours.

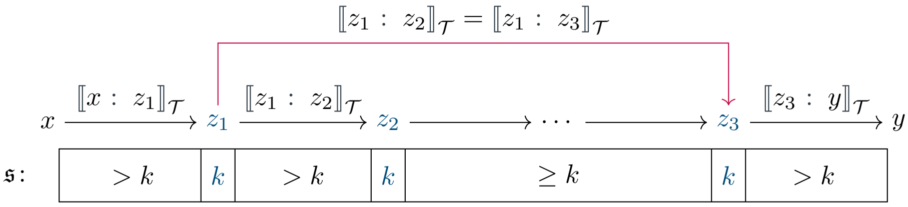

Figure 4 Fast computation of the value $[\![x : y]\!]_{\mathcal{T}}$ provided a forward Ramseyan split, using the fact that x and y are independent at level k.

A split $\mathfrak{s}$ is *forward Ramseyan* if for every $k \in \{1, \ldots, N\}$, every branch B of the tree, and every x, y, x', y' in the same k-neighbourhood of B, with $x \sqsubset_{\mathcal{T}} y$ and $x' \sqsubset_{\mathcal{T}} y'$,

$$[\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}} \cdot [\![x' : y']\!]_{\mathcal{T}} \quad . \tag{2}$$

In particular, $[\![x : y]\!]_{\mathcal{T}}$ is always an *idempotent*, that is, it satisfies $e \cdot e = e$. However, $[\![x : y]\!]_{\mathcal{T}}$ and $[\![x' : y']\!]_{\mathcal{T}}$ may be different idempotents.

Let us illustrate how one can use forward Ramseyan splits to compute efficiently the value of $[\![x : y]\!]_{\mathcal{T}}$ for any pair of nodes $x \sqsubset_{\mathcal{T}} y$. We say that x and y are *independent at level k* if $\mathfrak{s}(x : y) = k$ and there exist three nodes z_1, z_2, z_3 such that $x \sqsubseteq_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$, $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, and $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1 : z_2) > k$, and $\mathfrak{s}(z_3 : y) > k$. In this case, we can exploit forward Ramseyan splits to compute $[\![x : y]\!]_{\mathcal{T}}$ without inspecting the section between z_2 and z_3, leading to a fast (and first-order definable) computation. We refer to [7, Lemma 3] for further details.

The main theorem of [7] states that for every finite monoid M there exists a finite depth N such that for every tree $\mathcal{T}$ edge-labelled using M, there is a forward Ramseyan split of height N for $\mathcal{T}$. We can therefore assume that our trees are always equipped with a forward Ramseyan split of height N.

We now state the main remark that guides the rest of the paper: given a branch of a tree equipped with a forward Ramseyan split as in Figure 4, and assuming that $z_2 \neq z_3$, the value of $[\![x : y]\!]_{\mathcal{T}}$ does not change when inserting new nodes in the section between z_2 and z_3, provided these new nodes are assigned a split value at least k and the resulting tree remains forward Ramseyan. A fortiori, whether there is an edge between the leaves x and y in the resulting graph is not affected by such insertions. This observation guides our definition of a suitable ordering on trees in the upcoming Section 3.3.

3.3 Marked Nested Trees

In this section we show how the notion of forward Ramseyan split brings us closer to defining a well-quasi-order on trees that suits our purpose. A first step is to note that there already exists a notion of embedding that respects forward Ramseyan splits, namely gap-embeddings, which endow trees with a well-quasi-order [13].

▶ **Definition 11** ([13, Definition 3.3]). *A gap-embedding between trees $\mathcal{T}_1, \mathcal{T}_2$ endowed with forward Ramseyan splits $\mathfrak{s}_1, \mathfrak{s}_2$ is a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that*

1. *is a tree embedding: respects the ancestor relation $\sqsubseteq$, least common ancestors lca, and sibling ordering $\preceq$,*
2. *satisfies the root gap property: $\mathfrak{s}_2(r_2 \colon h(r_1)) \geq \mathfrak{s}_1(r_1)$ where r_1, r_2 are the roots of $\mathcal{T}_1, \mathcal{T}_2$,*
3. *satisfies the edge gap property: for every edge $x \sqsubset_{\mathcal{T}} y$ in $\mathcal{T}_1$, $\mathfrak{s}_2(h(x) \colon h(y)) \geq \mathfrak{s}_1(y)$.*

If the trees are node-labelled by a set $(X, \leq_X)$, we further require that for every node x in $\mathcal{T}_1$, the label of x is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$.

In Fact 12 we illustrate why gap-embeddings are well suited to our purpose: they respect the gaps induced by forward Ramseyan splits. Another key property of the gap-embedding ordering is that it is a well-quasi-ordering, as recalled in Theorem 13.

▶ **Fact 12.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be two trees and $\mathfrak{s}_1, \mathfrak{s}_2$ be two forward Ramseyan splits of height N of $\mathcal{T}_1, \mathcal{T}_2$ Assume that $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ is a gap-embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$. Then, for every u, v in $\mathcal{T}_1$, if $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \dots, N\}$, then $\mathfrak{s}_2(h(u) : h(v)) \geq k$.*

▶ **Theorem 13** ([13, Theorem 3.1, Main Theorem]). *The class of trees equipped with forward Ramseyan splits is well-quasi-ordered under the gap-embedding ordering, even under the assumption that nodes are labelled with a well-quasi-ordered set $(X, \leq_X)$.*

We now devise a modified version of the gap-embedding ordering to make the interpretation $\mathcal{I}$ order-preserving from trees to graphs. In particular, this modification will take into account the edge-labelling of the trees using the monoid M, that are not considered in the gap-embedding ordering.

▶ **Definition 14.** *A marked nested tree is a tuple $(\mathcal{T}, \mathfrak{s}, \rho)$ where $\mathcal{T}$ is a tree whose edges are labelled over a finite monoid M, $\mathfrak{s}$ is a forward Ramseyan split of height N of $\mathcal{T}$, and ρ is a function from the nodes of $\mathcal{T}$ to $\{\mathbf{M}, \mathbf{S}, \mathbf{D}\}$, respectively called marked, separating, and dummy nodes. A marked nested tree $(\mathcal{T}, \mathfrak{s}, \rho)$ is well-marked if the following conditions hold:*

1. *$\rho(\mathsf{root}) = \mathbf{M}$,*
2. *marked nodes are closed under least common ancestors,*
3. *for every pair of nodes $x \sqsubset_{\mathcal{T}} y$ such that x is marked, $\mathfrak{s}(x) = \mathfrak{s}(y) = k$ and $\mathfrak{s}(x : y) > k$, then y must be either marked or separating,*
4. *for every pair of nodes $x \sqsubset_{\mathcal{T}} y$ such that x is marked, $\mathfrak{s}(x) \neq \mathfrak{s}(y)$, $\mathfrak{s}(y) = k$ and $\mathfrak{s}(x : y) > k$, then y must be marked.*

This can be represented using the following figure

$$\begin{array}{ccccccccc} \mathbf{M} & & \mathbf{M} & & \in \{\mathbf{M}, \mathbf{S}\} & & \in \{\mathbf{M}, \mathbf{S}, \mathbf{D}\} & & \mathbf{M} \\ x & \xrightarrow[> k]{} & z_1 & \xrightarrow[> k]{} & z_2 & \xrightarrow[\geq k]{} & z_3 & \xrightarrow[> k]{} & y \\ & & k & & k & & k & & \end{array}$$

▶ **Definition 15.** *Given two marked nested trees $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, we say that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ gap-embeds into $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, written $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\mathrm{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, if there exists a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that*

- ⋆ *is a gap embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$,*
4. *sends the root of $\mathcal{T}_1$ to the root of $\mathcal{T}_2$,*
5. *maps leaves to leaves,*

6. *respects the marking:* $\rho_1 = \rho_2 \circ h$,
7. *respects local products: if y is the left (resp. right) child of x in $\mathcal{T}_1$, and y' is the left (resp. right) child of $h(x)$ in $\mathcal{T}_2$, then* $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : y']\!]_{\mathcal{T}_2}$.
8. *respects neighbourhood products: for every $k \in \{1, \ldots, N\}$, and every node $x \in \mathcal{T}_1$, if z is the closest ancestor of x such that $\mathfrak{s}_1(z) = k$ and z' is the closest ancestor of $h(x)$ such that $\mathfrak{s}_2(z') = k$, then* $[\![z : x]\!]_{\mathcal{T}_1} = [\![z' : h(x)]\!]_{\mathcal{T}_2}$.
9. *is gluing on non-dummy nodes: if $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and x is the parent of y in $\mathcal{T}_1$, then $h(x)$ is the parent of $h(y)$ in $\mathcal{T}_2$.*

Unlike the gap-embedding ordering of Theorem 13, the marked gap-embedding ordering of Definition 15 is not a well-quasi-order because of the gluing condition in Item 9. We can create an infinite antichain by considering the infinite sequence of trees $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ where $\mathcal{T}_n$ is a branch of length $n+1$, $\mathfrak{s}_n$ assigns the value 1 to every node of $\mathcal{T}_n$, and ρ_n marks all the nodes of $\mathcal{T}_n$ with $\mathbf{M}$. It is straightforward to check that for every $n < m$, $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ does not gap-embed into $(\mathcal{T}_m, \mathfrak{s}_m, \rho_m)$.

To recover the well-quasi-order property, we will restrict our attention to trees that make a limited use of the gluing property. An *L-bounded marked nested tree* is a marked nested tree such that any path of non-dummy nodes has length at most L. We prove in Theorem 16 that limiting the use of gluing restores the well-quasi-order property. Furthermore, we show in Lemma 17 that gap-embeddings between well-marked nested trees encode the composition ordering on trees when restricted to marked nodes. The combination of these two results in Corollary 19 is the main combinatorial ingredient of this section.

▶ **Theorem 16.** *For every finite L, the class of L-bounded well-marked nested trees labelled over a well-quasi-ordered set $(X, \leq_X)$ is well-quasi-ordered under the gap-embedding ordering.*

▶ **Lemma 17.** *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two marked nested trees, and assume that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\mathrm{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ via some mapping h. For every pair of marked nodes $x \sqsubset_{\mathcal{T}_1} y$ in $\mathcal{T}_1$, we have* $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.

Proof. We proceed by lexicographic induction on $(\mathfrak{s}_1(x\colon y),\ \ell(x,y))$, where $\ell(x,y)$ is the number of strict descendants on the path from x to y. If $\mathfrak{s}_1(x\colon y) = N+1$, then y is a child of x. Since y is marked, Item 9 yields that $h(y)$ is a child of $h(x)$, and then Item 7 gives $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$. For the induction steps, will use the following claim.

▷ Claim 18. Let $u \sqsubset_{\mathcal{T}_1} v$ with $\mathfrak{s}_1(v) = k$ and $\mathfrak{s}_1(u\colon v) > k$. If $\rho_1(v) \in \{\mathbf{M}, \mathbf{S}\}$, then $\mathfrak{s}_2(h(u)\colon h(v)) > k$.

Proof. The non-strict inequality $\mathfrak{s}_2(h(u)\colon h(v)) \geq k$ follows from Fact 12. Since v is non-dummy, Item 9 identifies the parent of $h(v)$ as the image of the parent of v, and therefore the first step above $h(v)$ cannot create a split value below or equal to k. Hence the inequality is strict. ◁

Assume that $k \triangleq \mathfrak{s}_1(x\colon y) \leq N$. Let z_1 be the first node on the path from x to y (strictly below x) such that $\mathfrak{s}_1(z_1) = k$. Then $\mathfrak{s}_1(x\colon z_1) > k$ and $\mathfrak{s}_1(z_1\colon y) \geq k$.

If $\mathfrak{s}(x) \neq k$. By Item 4, z_1 is marked (by Item 4), and we are in the following situation.

$$\begin{array}{ccccc} \mathbf{M} & & \mathbf{M} & & \mathbf{M} \\ x & \xrightarrow[> k]{\hspace{3em}} & z_1 & \xrightarrow[\geq k]{\hspace{3em}} & y \\ & & k & & k \end{array}$$

By Claim 18, we know that $\mathfrak{s}(h(x) : h(z_1)) > k$. Therefore, Item 8 ensures that $[\![x : z_1]\!]_{\mathcal{T}_1} = [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2}$. By induction hypothesis, because $\ell(z_1, y)$ is strictly smaller than $\ell(x, y)$ with the same split depth k, and both nodes are marked, we have $[\![z_1 : y]\!]_{\mathcal{T}_1} = [\![h(z_1) : h(y)]\!]_{\mathcal{T}_2}$. As a consequence, $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.

We can therefore without loss of generality assume that $\mathfrak{s}(x) = k$. Because $\mathcal{T}$ is well-marked, we know that z_1 is non-dummy by Item 3. We are therefore in the following situation:

$$\begin{array}{ccccc}
\mathbf{M} & & \in \{\mathbf{M}, \mathbf{S}\} & & \mathbf{M} \\
x & \xrightarrow[> k]{\qquad} & z_1 & \xrightarrow[\geq k]{\qquad} & y \\
k & & k & &
\end{array}$$

By Claim 18, we know that $\mathfrak{s}(h(x) : h(z_1)) > k$. Hence, Item 8 ensures that $[\![x : z_1]\!]_{\mathcal{T}_1} = [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2}$. Let z_2 be the first node below z_1 on the path to y with $\mathfrak{s}_1(z_2) = k$, and let z_3 be the first ancestor of $h(y)$ in $\mathcal{T}_2$ with split value k. By Claim 18 and Item 8, we obtain $[\![z_2 : y]\!]_{\mathcal{T}_1} = [\![z_3 : h(y)]\!]_{\mathcal{T}_2}$.

Since x and z_2 are in the same k-neighbourhood, and similarly $h(x)$ and z_3 are in the same k-neighbourhood, Equation (2) yields $[\![x : z_1]\!]_{\mathcal{T}_1} = [\![x : z_2]\!]_{\mathcal{T}_1}$, and $[\![h(x) : h(z_1)]\!]_{\mathcal{T}_2} = [\![h(x) : z_3]\!]_{\mathcal{T}_2}$. Therefore

$$\begin{aligned}
[\![x : y]\!]_{\mathcal{T}_1} &= [\![x : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_2 : y]\!]_{\mathcal{T}_1} \\
&= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_2 : y]\!]_{\mathcal{T}_1} \\
&= [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2} \cdot [\![z_3 : h(y)]\!]_{\mathcal{T}_2} \\
&= [\![h(x) : z_3]\!]_{\mathcal{T}_2} \cdot [\![z_3 : h(y)]\!]_{\mathcal{T}_2} \\
&= [\![h(x) : h(y)]\!]_{\mathcal{T}_2}.
\end{aligned}$$

This concludes all cases and the induction. ◀

▶ **Corollary 19.** *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two well-marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$. Then, the graph interpreted from $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is an induced subgraph of the graph interpreted from $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ when considering only marked leaves as vertices.*

Proof. Follows directly from Lemma 17 and the definition of the monoid interpretation using the set $P \subseteq M^3$ (see Equation (1)): edges between two marked leaves x and y only depend on the values of $[\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_{\mathcal{T}}$, $[\![\mathsf{lca}(x, y) : x]\!]_{\mathcal{T}}$, and $[\![\mathsf{lca}(x, y) : y]\!]_{\mathcal{T}}$, all of which are marked nodes because the trees are well-marked. ◀

4 Bad Patterns in Images of Interpretations

In this section we fix a monoid interpretation $\mathcal{I} = (\Sigma, \mu, M, P)$ from trees edge-labelled over a finite alphabet Σ to finite undirected graphs. We furthermore assume that all trees we consider are equipped with a forward Ramseyan split $\mathfrak{s}$ over the finite monoid M. Our goal is to isolate some combinatorial obstructions to being 2-well-quasi-ordered in classes of graphs that are images of monoid interpretations from trees. The key idea of this section is to start from a tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, and try to understand under which conditions one can turn this tree into a L-bounded marked nested tree representing the same graph, allowing us to leverage Theorem 16 to conclude that the class of graphs is $\forall$-well-quasi-ordered.

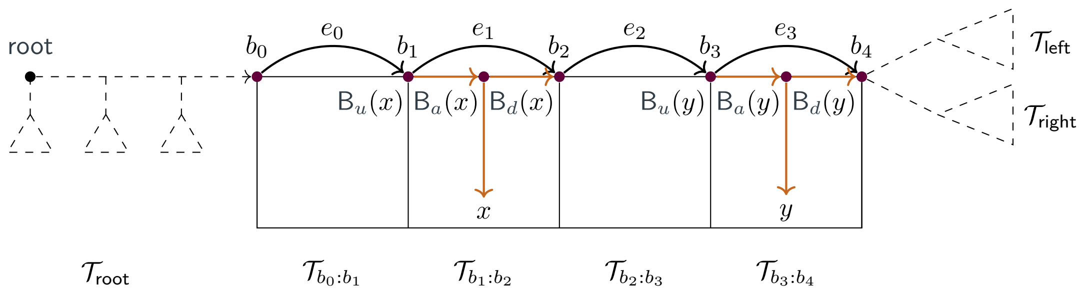

Figure 5 Partitioning a branch of a tree $\mathcal{T}$ using a bough B. Here, the nodes b_i for $0 \leq i \leq 4$ are k-neighbours forming the backbone B. The letter e_i denotes idempotent monoid element $[\![b_i : b_{i+1}]\!]_{\mathcal{T}}$. In dashed, we represented the context $C[\square]$ such that $\mathcal{T} = C[B]$. To compute the presence of an edge between x and y in the resulting graph, it is sufficient to know the values of $[\![b_2 : b_3]\!]_{\mathcal{T}}$, and the respective bough types of x and y in their respective bough blocks $\mathcal{T}_{b_1:b_2}$ and $\mathcal{T}_{b_3:b_4}$.

▶ **Definition 20.** *Let $\mathcal{T}$ be a tree and $\mathfrak{s}$ be a forward Ramseyan split of height N. Let $n \geq 1$, and let $b_0, b_1, \ldots, b_n$ be distinct nodes on a branch of $\mathcal{T}$ such that $b_0 \sqsubseteq b_1 \sqsubseteq \cdots \sqsubseteq b_n$, and for every $0 \leq i < n$, $\mathfrak{s}(b_i) = k$ and $\mathfrak{s}(b_i \colon b_{i+1}) > k$. The bough of level k with backbone $b_0, b_1, \ldots, b_n$ is the subtree of $\mathcal{T}$ induced by all nodes x such that $b_0 \sqsubseteq x$ and $\neg(b_n \sqsubseteq x)$, plus b_n itself. We say that the dimension of the bough B is n.*

Let B be a bough of level k in $\mathcal{T}$ and let $b_0, b_1, \ldots, b_n$ be its backbone. Given indices $i < j$, we define the $\mathcal{T}_{b_i:b_j}$ as the set of nodes x in $\mathcal{T}$ such that $b_i \sqsubseteq x$ and $\neg(b_j \sqsubseteq x)$. When $j = i + 1$, this set of nodes is called a *block*. We also refer to Figure 5 for an illustration of the resulting partition of the tree $\mathcal{T}$ with respect to a given bough B.

For every leaf x of $\mathcal{T}$ that is in $\mathcal{T}_{b_0:b_n}$, we define $\mathsf{B}_a(x) \triangleq \mathsf{lca}(x, b_n)$. Moreover, we define $\mathsf{B}_u(x)$ (resp. $\mathsf{B}_d(x)$) to be the maximal (resp. minimal) element of the backbone $b_0, b_1, \ldots, b_n$ such that $\mathsf{B}_u(x) \sqsubseteq \mathsf{B}_a(x) \sqsubseteq \mathsf{B}_d(x)$. Using these reference points, we define the following tuple monoid elements associated to a leaf x in a block of B:

$$([\![\mathsf{B}_a(x) : x]\!]_{\mathcal{T}}, [\![\mathsf{B}_u(x) : \mathsf{B}_a(x)]\!]_{\mathcal{T}}, [\![\mathsf{B}_a(x) : \mathsf{B}_d(x)]\!]_{\mathcal{T}}, [\![\mathsf{root} : \mathsf{B}_u(x)]\!]_{\mathcal{T}}).$$

We call this tuple the *bough type* of the leaf x with respect to the bough B. We again refer to Figure 5 for an illustration of the type of a leaf with respect to a given bough B. There are only finitely many possible bough types because M is finite.

Throughout the paper, we often rely on tree surgery operations consisting of replacing a bough in a tree by another bough. Given a bough B in a tree $\mathcal{T}$ defined using a backbone $b_0, b_1, \ldots, b_n$, we write $\mathcal{T} = C[B]$ to denote that $\mathcal{T}$ is obtained by plugging the bough B into a *context* $C[\square]$. Formally, a *context* $C[\square]$ is given by a tree $\mathcal{T}_{\mathsf{root}}$ with a distinguished leaf $\square$, together with two trees $\mathcal{T}_{\mathsf{left}}$ and $\mathcal{T}_{\mathsf{right}}$, and two monoid elements m_{left} and m_{right}. These elements are represented as dashed lines in Figure 5. The tree $C[B]$ is then obtained as follows: one first attaches $\mathcal{T}_{\mathsf{left}}$ as the left subtree of b_n with an edge labelled m_{left}, then attaches $\mathcal{T}_{\mathsf{right}}$ as the right subtree of b_n with an edge labelled m_{right}, and finally replaces the distinguished leaf $\square$ in $\mathcal{T}_{\mathsf{root}}$ with the node b_0.

In order to simplify statements, we will say that the *leaves* of a context $C[\square]$ are the leaves of the tree $\mathcal{T}_{\mathsf{root}}$, $\mathcal{T}_{\mathsf{left}}$, and $\mathcal{T}_{\mathsf{right}}$, excluding the distinguished leaf $\square$. As a consequence, the leaves of $C[B]$ are exactly the leaves of $C[\square]$ together with the leaves of the blocks of B. Furthermore, we will often need to talk about the *first idempotent value* of a bough B

defined using the backbone $b_0, b_1, \ldots, b_n$, which is the idempotent element $[\![b_0 : b_1]\!]_{\mathcal{T}}$.[3]

Given two boughs B and B' of level k, and a context $C[\square]$, one can consider the trees $C[B]$ and $C[B']$. These trees have edges labelled over M and are equipped with splits induced by the splits on $C[\square]$ and B (resp. B'). We say that B and B' are *compatible* if they share the same first idempotent value and if for every context $C[\square]$ such that the tree $C[B]$ (with split $\mathfrak{s}$) is forward Ramseyan, the tree $C[B']$ (with split $\mathfrak{s}'$) is also forward Ramseyan.

▶ **Definition 21.** *Let B be a bough of level k in a tree $\mathcal{T} = C[B]$. We say that B is a good bough if there exists a compatible bough H of level k and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ such that:*

1. *h is an embedding of graphs,*
2. *h is the identity map on leaves of $C[\square]$,*
3. *there exist three consecutive blocks in H whose leaves are left untouched*[4] *by h.*

A bad bough is a bough that is not a good bough.

We claim that the existence of bad boughs of arbitrarily large dimension is the only obstruction to being 2-well-quasi-ordered.

▶ **Lemma 22.** *The following are equivalent:*

1. *The image of $\mathcal{I}$ is 2-well-quasi-ordered.*
2. *The image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.*
3. *There exists a bound on the dimension of bad boughs.*

It is clear that Item 2 implies Item 1. In Section 4.1 we will prove that Item 3 implies Item 2, while in Section 4.2 we will prove that Item 1 implies Item 3, concluding the proof of Lemma 22.

4.1 Using the Gap Embedding

In this section, we leverage Theorem 16 to prove that Item 3 implies Item 2 of Lemma 22. The main idea is to use the bound on the dimension of bad boughs to insert "dummy nodes" inside our trees, turning them into marked nested trees. Because we can insert these dummy nodes in a controlled manner, we are able to ensure that the resulting marked nested trees are L-bounded for some $L \in \mathbb{N}$, allowing us to apply Theorem 16.

▶ **Lemma 23.** *Assume that there exists a bound N_0 on the dimension of bad boughs. Then, for every tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, one can effectively construct a well-marked nested tree $(\mathcal{T}', \mathfrak{s}', \rho)$ that is L-bounded for some $L \in \mathbb{N}$, and such that $\mathcal{I}(\mathcal{T})$ is an induced subgraph of $\mathcal{I}(\mathcal{T}')$, when the latter is restricted to vertices corresponding to marked leaves.*

Proof. We proceed iteratively, starting from the tree $\mathcal{T}$ where every node is set as a marked node. A rewrite step works as follows: given a bough backbone $b_0, b_1, \ldots, b_n$ defining a bough B of level k where all nodes of the bough backbone are marked nodes and $n > N_0$, we know that B is a good bough by assumption, and we write $\mathcal{T} = C[B]$.

By definition of good boughs, there exists a compatible bough H and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ satisfying the three conditions of Definition 21. Let us define $\mathcal{T}' = C[H]$, which is

[3] It is also the element $[\![b_0 : b_n]\!]_{\mathcal{T}}$, thanks to the forward Ramseyan property of $\mathfrak{s}$.
[4] Here we say that a leaf of H is untouched if the corresponding vertex of $\mathcal{I}(C[H])$ is not in the image of h.

still forward Ramseyan by definition of compatible boughs. Let us now define the vertex-labelling of $\mathcal{T}'$. For nodes of $C[\Box]$, we keep the same vertex-labelling as in $\mathcal{T}$. For nodes of H, we identify the three consecutive blocks whose leaves are left untouched by h as $H_{z_i:z_{i+1}}$, $H_{z_{i+1}:z_{i+2}}$, and $H_{z_{i+2}:z_{i+3}}$. We label all these nodes as dummy nodes except z_i and z_{i+3}, which are labelled as marked nodes, and z_{i+1}, which is labelled as a separating node. Finally, all other nodes of H are labelled as marked nodes.

We claim that one can apply these rewrite steps as follows: first, apply them to boughs of minimal split level until exhaustion, and then proceed with the next split level, and so on. To a tree $\mathcal{T}$ and a split level k, one can associate the multiset of dimensions of all boughs of level k with backbone made of marked nodes inside $\mathcal{T}$. Since each rewrite step decreases this multiset for the Dershowitz–Manna ordering this process terminates [14].

Let us first observe that every rewrite step preserves the fact that $\mathcal{T}$ is a well-marked nested tree. This is because non-marked nodes are only inserted after a separating node.

Let us now prove that every rewrite step preserves the fact that the graph $\mathcal{I}(\mathcal{T})$ restricted to vertices corresponding to marked leaves is an induced subgraph of $\mathcal{I}(\mathcal{T}')$ restricted to vertices corresponding to marked leaves. This follows directly from the fact that $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ is an embedding of graphs and that h is the identity on leaves of $C[\Box]$.

From our construction, it is clear that at every level k, there are at most N_0+1 consecutive marked nodes in the bough backbone of any bough of level k.

Let us now prove that the resulting tree $\mathcal{T}'$ is L-bounded for some $L \in \mathbb{N}$. Assume towards a contradiction that there exists arbitrarily long paths of non-dummy nodes in the resulting tree $\mathcal{T}'$. To a pair of nodes $x \sqsubseteq y$ in such a path, we associate the colour $(\mathfrak{s}'(x), \mathfrak{s}'(x\colon y), \mathfrak{s}'(y))$. By a Ramsey argument, if the path is too long, there exists a sequence $x_0 \sqsubseteq x_1 \sqsubseteq \cdots \sqsubseteq x_{N_0+1}$ of non-dummy nodes in the path such that $\mathfrak{s}'(x_i) = \mathfrak{s}'(x_j)$ and $\mathfrak{s}'(x_i\colon x_j) = \mathfrak{s}'(x_{i'}\colon x_{j'})$ for every $0 \leq i < j \leq N_0+1$ and $0 \leq i' < j' \leq N_0+1$. Without loss of generality, we can assume that $\mathfrak{s}'(x_i) < \mathfrak{s}'(x_i\colon x_{i+1})$ for every $0 \leq i \leq N_0$. Since we only insert separating node prior to a dummy node, we can also assume that all nodes $x_0, x_1, \ldots, x_{N_0+1}$ are marked nodes. We have identified a bough backbone $x_0, x_1, \ldots, x_{N_0+1}$ of marked nodes defining a bough of level $\mathfrak{s}'(x_0)$, which contradicts the fact that the rewriting procedure terminated. ◀

Proof of Item 3 implies Item 2 of Lemma 22. Let us assume that there exists a bound on the dimension of bad boughs, and let us consider a well-quasi-ordered set $(Y, \leq)$ of labels. Let $(G_i)_{i\in\mathbb{N}}$ be an infinite sequence of graphs in $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_\Sigma))$. We want to show that there exist two indices $i < j$ such that G_i is a labelled induced subgraph of G_j.

First, every graph G_i can be represented as $\mathcal{I}(\mathcal{T}_i)$ for some tree $\mathcal{T}_i$ equipped with a forward Ramseyan split $\mathfrak{s}_i$. Then, by Lemma 23, there exists a finite $L \in \mathbb{N}$ such that we can construct L-bounded well-marked nested tree $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i)$ such that G_i is an induced subgraph of $\mathcal{I}(\mathcal{T}_i')$, when the latter is restricted to vertices corresponding to marked leaves. By labelling the marked leaves of $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i)$ with $\circ$ if they are not in G_i, and with their original label in Y otherwise, we obtain a sequence of $Y \cup \{\circ\}$ node-labelled well-marked nested trees.

By Theorem 16, there exist two trees $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i')$ and $(\mathcal{T}_j', \mathfrak{s}_j', \rho_j')$ in this sequence such that $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i) \leq_{\mathrm{gap}} (\mathcal{T}_j, \mathfrak{s}_j, \rho_j)$. By Corollary 19, the graph interpreted from $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i)$ is a labelled induced subgraph of the graph interpreted from $(\mathcal{T}_j', \mathfrak{s}_j', \rho_j')$ when both are restricted to marked leaves. We conclude that G_i is a labelled induced subgraph of G_j, and thus $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_\Sigma))$ is well-quasi-ordered. Therefore, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered. ◀

4.2 Obstructions

In this section we prove that Item 1 implies Item 3 of Lemma 22. We proceed by contraposition and assume that there exist bad boughs of arbitrarily large dimension. We use this to construct an infinite antichain in the image of $\mathcal{I}$ using two colours. To build such an infinite antichain, we extract from an infinite sequence of bad boughs of increasing dimension some incomparable graphs. We first show that all these bad boughs can be assumed to be compatible (see Lemma 24 below), and that we can place all these bad boughs in the *same* context $C[\square]$ to witness that they are bad (see Lemma 25 below).

▶ **Lemma 24.** *Let k be a level. There are finitely many boughs of level k up to compatibility.*

▶ **Lemma 25.** *There exists a finite set $\mathcal{F}$ of contexts $C[\square]$ such that for every bad bough B of level k, there exists a context $C[\square] \in \mathcal{F}$ such that B is a bad bough in $C[B]$.*

Proof. The proof relies on the fact that the context $C[\square]$ plays a very little role in witnessing that a bough B is bad.

Let B and H be two compatible boughs of level k, and let $C[\square]$ be a context such that $\mathcal{T} = C[B]$ and $\mathcal{T}' = C[H]$ are forward Ramseyan. We want to characterise when there exists an embedding $h\colon \mathcal{I}(\mathcal{T}) \to \mathcal{I}(\mathcal{T}')$ witnessing that B is a good bough in $C[B]$ with respect to H.

First, we only need to consider edges between leaves of B or edges between leaves of B and leaves of $C[\square]$. Indeed, the map h acts as the identity on leaves of $C[\square]$, so edges are preserved there.

The presence of edges between leaves of B depends only on their respective bough types and the value of $[\![\mathsf{root} : \square]\!]_{\mathcal{T}_{\mathsf{root}}}$. Consequently, any two contexts sharing the same value for the path from the root to $\square$ do not change the edges between leaves of B (or H). Since there are finitely many monoid elements, there are finitely many possible values for this path.

Let us now consider edges between leaves of B and leaves of $C[\square]$. Similarly as for the leaves in boughs, one can associate to every leaf of $C[\square]$ a *context type* allowing to determine the presence of an edge. For a leaf x in $C[\square]$ that belongs to the root subtree, we collect the following monoid elements: $[\![\mathsf{root} : z]\!]_{\mathcal{T}_{\mathsf{root}}}$, $[\![z : x]\!]_{\mathcal{T}_{\mathsf{root}}}$, and $[\![z : \square]\!]_{\mathcal{T}_{\mathsf{root}}}$, where z is the closest ancestor of x belonging to the path from root to $\square$. For a leaf that belongs to the left (resp. right) subtree, we only consider the monoid element $m_{\mathsf{left}} \cdot [\![\mathsf{root} : x]\!]_{\mathcal{T}_{\mathsf{left}}}$ (resp. $m_{\mathsf{right}} \cdot [\![\mathsf{root} : x]\!]_{\mathcal{T}_{\mathsf{right}}}$). It is immediate that to compute the presence of an edge between a leaf y in B and a leaf x in $C[\square]$, it suffices to know the bough type of y and the context type of x (see Equation (1), and Figure 5). Since there are finitely many possible context types and bough types, if a bough B is a good bough in $C[B]$ using a map h and a compatible bough H, then for any other context $C'[\square]$ sharing the same context types for its leaves as $C[\square]$, the bough B is also a good bough in $C'[B]$ using the same map h and the same compatible bough H.

Since there are finitely many possible contexts up to this equivalence relation, we conclude the proof. ◀

Proof of Item 1 implies Item 3 of Lemma 22. Assume towards a contradiction that the class of graphs obtained as images of $\mathcal{I}$ is 2-well-quasi-ordered, but that there exist bad boughs of arbitrarily large dimension. Let $(B_i)_{i\geq 1}$ be an infinite sequence of such bad boughs. Without loss of generality, we can assume that all boughs B_i have the same level k and are pairwise compatible, by Lemma 24.

Using Lemma 25, we can extract one context $C[\square]$ such that infinitely many boughs B_i are bad boughs in $C[B_i]$. Finally, we can extract further to assume that the dimension of the boughs B_{i+1} is greater than thrice the number of leaves in B_i.

Let us now colour every graph $\mathcal{I}(C[B_i])$ with two labels: $\bullet$ to leaves belonging to B_i, and colour $\circ$ to leaves belonging to $C[\square]$. Because we assume that the image of $\mathcal{I}$ is 2-well-quasi-ordered, we can extract an infinite increasing subsequence, and assume that for all $i < j$, there exists an embedding $f_{i,j} \colon \mathcal{I}(C[B_i]) \to \mathcal{I}(C[B_j])$ that preserves labels. Let us remark that for all $i < j$, the map $f_{i,j}$ acts as a permutation when restricted to nodes labelled $\circ$, that is of size the number of leaves in $C[\square]$. By a Ramsey argument, we can therefore assume that this permutation is the identity for all $i < j$.

Finally, consider any two indices $i < j$. Because the dimension of B_j is greater than thrice the number of leaves in B_i, there must exist three consecutive bough blocks of B_j that are left untouched by $f_{i,j}$. Furthermore, we know that $f_{i,j}$ is the identity on leaves belonging to $C[\square]$. Hence, the map $f_{i,j}$ witnesses that B_i is a good bough in $C[B_i]$, which is a contradiction. ◀

5 Bounded Clique Width

In this section, we will aim at leveraging the previous results to tackle classes of bounded clique-width. Our first goal is to prove our Theorem 4 regarding the images of finite trees under MSO-interpretations. There are two difficulties here: first, we have only dealt with simple MSO-interpretations so far, and second, we have not yet given any decision procedure to check whether the image of a given simple MSO-interpretation is 2-well-quasi-ordered. We first address the second difficulty.

▶ **Definition 26** (Perfect Bough). *Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k. We say that B is a perfect bough if there exists a compatible bough H of level k and a map $h \colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$*

1. *h is an embedding of graphs,*
2. *h is the identity map on leaves of $C[\square]$,*
3. *the bough type of every leaf x in B is the same as the bough type of $h(x)$ in H,*
4. *there exist three consecutive blocks in H whose leaves are left untouched by h.*

An imperfect bough is a bough that is not a perfect bough.

Note that every perfect bough is a good bough, thus every bad bough is an imperfect bough. This allows us to lift Lemma 22 to perfect boughs.

▶ **Lemma 27.** *The following are equivalent:*

1. *the image of $\mathcal{I}$ is 2-well-quasi-ordered,*
2. *the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered,*
3. *there exists a bound on the dimension of imperfect boughs.*

Proof. First, if we have a bound on the dimension of imperfect boughs, then we also have a bound on the dimension of bad boughs. Hence, by Lemma 22, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.

Conversely, assume that there exist imperfect boughs of arbitrarily large dimension. Then, by an argument similar to the one of Section 4.2, we can construct an infinite labelled antichain in the image of $\mathcal{I}$. The only change is that one also adds the bough type of the leaves as additional labels. This actually follows the original proof technique of [23] for bounded linear clique-width. ◀

The important advantage of Lemma 27 over Lemma 22 is that imperfect boughs are easily recognizable using Monadic Second-Order logic on trees. Indeed, every bough B is actually a (binary) tree with some distinguished nodes that form its backbone. Hence, given a tree $\mathcal{T}$ together with a split $\mathfrak{s}$ and labels on its edges coming from a finite monoid M, one can ask whether this tree with distinguished nodes represents an imperfect bough.

In order to do so, we first need the following Fact 28, which explains that when replacing a bough B by a compatible bough H, we do not change the part of the graph represented by the leaves of the bough context $C[\Box]$. This is because, to determine the presence of edges between leaves of $C[\Box]$, one needs only to know the value of the first idempotent element labelling the bough, which does not change when replacing B by a compatible bough H. Consequently, we need focus only on the leaves of B inside of H to construct an embedding from $\mathcal{I}(C[B])$ to $\mathcal{I}(C[H])$ satisfying the conditions of Definition 26.

▶ **Fact 28.** *Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k, and let H be a bough that is compatible with B. Let $\mathcal{T}' = C[H]$ be the tree obtained by replacing B by H in $\mathcal{T}$. Then, the subgraph of $\mathcal{I}(\mathcal{T})$ induced by the leaves in the context $C[\Box]$ equals the subgraph of $\mathcal{I}(\mathcal{T}')$ induced by the leaves in the context $C[\Box]$.*

▶ **Lemma 29.** *There exists an* MSO *formula that recognizes imperfect boughs.*

▶ **Corollary 30.** *We can decide whether there are imperfect boughs of arbitrarily large dimension.*

Proof. We use the theory of Cost-MSO formulas [8, Section 6]. Let Φ denote the formula of Lemma 29 that recognises imperfect boughs. We now consider the following Cost MSO formula:

$$\Psi(N) \triangleq \Phi \wedge |\{ \text{ dimension of the bough } \}| \geq N$$

This formula takes an input parameter N (a natural number) and decides whether the provided tree with split and labels is an imperfect bough of dimension at most N. Because we can compare the boundedness of Cost-MSO definable functions [8, Corollary 16], we can decide whether such a formula can hold for arbitrarily large values of N, effectively answering our question. ◀

We are now ready to prove our main result.

▶ **Theorem 4.** *Let $\mathcal{C}$ be the image of a class of finite trees under an* MSO *interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*
1. *$\mathcal{C}$ is 2-well-quasi-ordered,*
2. *$\mathcal{C}$ is $\forall$-well-quasi-ordered,*
3. *$\mathcal{C}$ is orderably $\forall$-well-quasi-ordered,*

Proof of Theorem 4. Let $\mathcal{I}$ be an MSO-interpretation. We can eliminate the domain formula and selection formula by standard arguments without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$ [23, Lemma 8]. Thus, we can assume that $\mathcal{I}$ is a simple MSO-interpretation. Furthermore, the transformation from a simple MSO-interpretation to a monoid interpretation from trees described in Section 3 is effective. Finally, by Lemma 27 the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if it is $\forall$-well-quasi-ordered, and by Lemma 29, we can decide whether the image of $\mathcal{I}$ is 2-well-quasi-ordered.

It remains to discuss the fact that we can add a total ordering on the graphs without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$. This is because we prove that the image of $\mathcal{I}$ is 2-well-quasi-ordered using the marked gap-embedding ordering on marked nested trees representing the graphs, which already includes a total ordering on the leaves. ◀

It is straightforward to derive that the same holds for hereditary classes of bounded clique-width, thus proving Theorem 5.

▶ **Theorem 5.** *The conclusion of Theorem 4 holds when $\mathcal{C}$ is a hereditary class of graphs having bounded clique-width.*

Proof of Theorem 5. By a standard argument, if the class $\mathcal{C}$ is hereditary and 2-well-quasi-ordered, then it is characterised by finitely many forbidden induced subgraphs [11, Proposition 3]. Thus, if $\mathcal{C}$ has bounded clique-width and is hereditary, we can assume that $\mathcal{C}$ is exactly the image of an MSO-interpretation $\mathcal{I}$ from finite trees: this is done by ensuring in the formula ϕ_{dom} that the graph that would be produced does not contain any of these forbidden induced subgraphs. The result then follows from Theorem 4. ◀

Let us now turn to Theorem 6, and in particular the link between 2-well-quasi-ordering and bounded linear clique-width. The following lemma shows that the existence of imperfect boughs of arbitrarily large dimension can be witnessed using only blocks from a finite set.

▶ **Lemma 31.** *Assume that there are imperfect boughs of arbitrarily large dimension. Then there exists a finite set of blocks $\mathbb{F}$ such that we can build an imperfect bough of arbitrarily large dimension using only blocks from $\mathbb{F}$.*

As an immediate consequence of Lemma 27 and Lemma 31, we can transform long imperfect boughs into long imperfect boughs using only blocks from a finite set. This allows us to prove the following corollary.

▶ **Corollary 32.** *The image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if every subclass of bounded linear clique-width in $\mathcal{I}$ is 2-well-quasi-ordered.*

Proof. If the image of $\mathcal{I}$ is 2-well-quasi-ordered, then every subclass of bounded linear clique-width in $\mathcal{I}$ is also 2-well-quasi-ordered. Conversely, if the image of $\mathcal{I}$ is not 2-well-quasi-ordered, then by Lemma 27 there exist imperfect boughs of arbitrarily large dimension. We consider the finite set of blocks $\mathbb{F}$ given by Lemma 31, to obtain an infinite sequence of imperfect boughs $(B_i)_{i\in\mathbb{N}}$ of increasing dimension using only blocks from $\mathbb{F}$.

Note that if B is an imperfect bough in some tree $\mathcal{T} = C[B]$, then it remains an imperfect bough in any context $C'[\square]$, as the definition of perfect bough is local to the bough itself. Thus, we can select a single context $C[\square]$ such that B_i is an imperfect bough inside of $C[\square]$ for all $i \in \mathbb{N}$. Similarly as in Section 4.2, we can then build an infinite antichain of two-labelled graphs inside of $\mathcal{I}$ by colouring vertices of $\mathcal{I}(C[B_i])$ with $\circ$ if they come from $C[\square]$ and with $\bullet$ if they come from B_i.

Finally, note that the collection of graphs $\mathcal{I}(C[B_i])$ for $i \in \mathbb{N}$ has bounded linear clique-width. Indeed, the context $C[\square]$ is fixed, and the blocks used to build the boughs B_i come from a finite set $\mathbb{F}$. Consequently, we can represent these graphs by using one colour per node of the context $C[\square]$ plus one colour per node in each block in $\mathbb{F}$, and write single branch (linear) expressions representing all graphs $\mathcal{I}(C[B_i])$. ◀

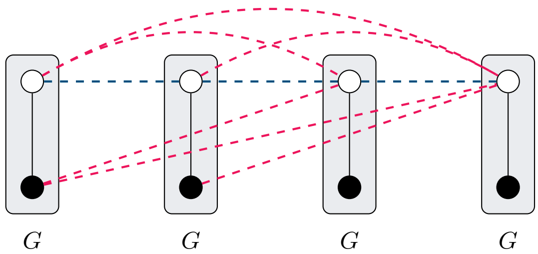

Figure 6 The graph S_4 as described in Example 34. The dashed edges represents the edges defined by C and F, respectively in blue and in red. Vertices originating from the same copy of G are grouped in rectangles.

6 Structural Characterisations

In this section, we will be interested in extracting a very regular structure from a graph class $\mathcal{C}$ that is not $\forall$-well-quasi-ordered. This structure will take three forms, the first one is a combinatorial notion of regular sequence, in continuation with the work of [23] and our Corollary 32. The second one is a notion of periodic sequence that originates from [4, Section 7], and is at the heart of their [4, Conjecture 2]. Those structural notions encode "path-like" behaviour, and we actually manage to extract from those all finite paths using an existential transduction, answering positively to Conjecture 3.

▶ **Definition 33.** *A regular sequence (of graphs) is given by a finite graph G, a labelling function $\ell \colon V(G) \to \Sigma$ for some finite set Σ, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G^r)_{r \geq 1}$ as follows: the vertex set of G^r is $V(G) \times \{1, \dots, r\}$, and there is an edge between (u, i) and (v, j) with $i \leq j$ if and only if one of the following holds:*

- *$(u, v) \in E(G)$ and $i = j$;*
- *$(\ell(u), \ell(v)) \in C$ and $|i - j| = 1$*
- *$(\ell(u), \ell(v)) \in F$ and $|i - j| > 1$.*

▶ **Example 34.** Let us consider the graph G that is a path with two vertices, respectively labelled $\bullet$ and $\circ$. Let us define $C \triangleq \{(\circ, \circ)\}$ and $F \triangleq \{(\circ, \circ), (\bullet, \circ)\}$, the graph G^n is the *split-permutation graph of size n* that we write S_n and can be seen in Figure 6.

A *regular antichain* is a regular sequence that is an antichain for the labelled induced subgraph relation when vertices of G^r are labelled by their color in G, and the first and last copies of G in G^r are additionally labelled by two distinct new labels $\triangleright$ and $\triangleleft$. It follows results from [23] and our previous Section 5 that such structures can be found in non-$\forall$-well-quasi-ordered hereditary classes of graphs of bounded clique-width.

▶ **Lemma 35.** *Let $\mathcal{C}$ be a hereditary class of graphs of bounded clique-width that is not $\forall$-well-quasi-ordered. Then, it contains a regular antichain.*

Our next goal is to show that can existentially transduce all finite paths from a regular antichain. Remark that split-permutation graphs form a regular antichain, but exclude P_4 as an induced subgraph. Before that, let us briefly recall that existentially transducing all paths is an obstruction to being $\forall$-well-quasi-ordered.

▶ **Lemma 36.** *Let $\mathcal{C}$ be a hereditary class that existentially transduces all paths, then $\mathcal{C}$ is not $\forall$-well-quasi-ordered.*

Proof. Let $\mathcal{I} = (\phi_{\text{univ}}, \phi_{\text{edge}}, \phi_{\text{dom}})$ be an existential interpretation from $\mathsf{Label}_\Sigma(\mathcal{C})$ to the class of finite graphs, such that the image of $\mathsf{Label}_\Sigma(\mathcal{C})$ contains all finite paths. Let us write G_i for a graph in $\mathsf{Label}_\Sigma(\mathcal{C})$ such that $\mathcal{I}(G_i) = P_i$, where P_i is the path on i vertices. Remark that if $G_i \subseteq_i G_j$ as labelled graphs, then $P_i = \mathcal{I}(G_i)$ is a subgraph (not necessarily induced) of $\mathcal{I}(G_j) = P_j$. Indeed, whenever $\phi_{\text{univ}}(u)$ holds in G_i, it also holds in G_j, because G_i is an induced subgraph of G_j and ϕ_{univ} is existential. Similarly, whenever $\phi_{\text{edge}}(u, v)$ holds in G_i, it also holds in G_j.

Let us consider an extra labelling information on the vertices of the graphs in $\mathsf{Label}_\Sigma(\mathcal{C})$, where we color the two endpoints of each path P_n with two distinct colors $\triangleright$ and $\triangleleft$. Using these new labels, one can existentially interpret the class of all finite paths with endpoints colored with $\triangleright$ and $\triangleleft$.

Assume towards a contradiction that $\mathcal{C}$ is $\forall$-well-quasi-ordered. Then, there is a pair $i < j$ such that $G_i \subseteq_i G_j$ as labelled graphs. By the previous remark, this implies that P_i (with coloured endpoints) is a labelled subgraph of P_j (with coloured endpoints), which is impossible. ◀

▶ **Lemma 37.** *Let $\mathcal{C}$ be a regular antichain. Then $\mathcal{C}$ existentially transduces the class of all finite paths.*

Proof sketch. Let $\mathcal{C} = (G^r)_{r \geq 1}$ and let $\ell \colon V(G) \to \Sigma$ be its labelling function with sets $C, F \subseteq \Sigma^2$. Without loss of generality, assume that ℓ is injective up to duplicating labels from Σ, and assume that $|\Sigma| = |V(G)|$. We define the following quantifier-free formula that uses the edge predicate on G^r and the labelling function $\ell \colon V(G^r) \to \Sigma$.

$$\phi_{\rightarrow}(u, v) = \neg(E(u, v) \Leftrightarrow (\ell(u), \ell(v)) \in F) \quad .$$

Informally, $\phi_{\rightarrow}(u, v)$ creates an arc (u, v) if the edge relation between $u, v \in G^r$ differs if u and v are put far away form each other in this order.[5]

To obtain a path, we first show that one can obtain a long directed shortest path from the first copy of G to the last copy of G, going through every copy of G. Then, from this so-called *spanning path*, we extract undirected paths by analysing the behaviour of potential “backward edges” along the directed path using existential formulas built upon $\phi_{\rightarrow}$. ◀

Finally we provide a last structural characterisation defined in [4, Section 7] that is not $\forall$-well-quasi-ordered similar to periodic sequence of graphs.

▶ **Definition 38** ([4, Section 7])**.** *A periodic sequence (of graphs) is given by a finite word w over a finite alphabet Σ, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G_{w^r})_{r \geq 1}$ as follows: the vertex set of G_{w^r} is $\{1, \ldots, r \cdot |w|\}$, and there is an edge between $i \leq j$ if*

- *$i + 1 = j$ and $((w^r)_i, (w^r)_j) \in C$,*
- *$i + 1 < j$ and $((w^r)_i, (w^r)_j) \in F$.*

▶ **Example 39.** Let us consider the word $w = \circ\bullet$, and $C \triangleq \{(\circ, \bullet)\}$ and $F \triangleq \{(\circ, \circ), (\bullet, \circ)\}$. The graph G_{w^n} is the split-permutation graph of size n.

It is immediate to see that periodic sequences are a special case of regular sequences, by grouping vertices of w into a single graph G. A *periodic antichain* is a periodic sequence that is an antichain for the labelled induced subgraph relation when vertices of G_{w^r} are not labelled, except for the first and last vertices respectively labelled by $\triangleright$ and $\triangleleft$. In the original

[5] It could be that u can be in a copy of G that comes after the one of v in G^r.

definition of [4], it was required by periodic sequences to have $C = \Sigma^2 \setminus F$. It is easy to verify that with this condition a periodic sequence is an antichain if the first and last vertices are labelled.

While it is not immediate from the definitions that one can turn a regular antichain into a periodic antichain, it follows from our proof of Lemma 37 that when we existentially transduce all paths from a regular antichain, we actually isolate a periodic antichain.

▶ **Corollary 40.** *Let $\mathcal{C}$ be a hereditary class, then $\mathcal{C}$ contains a regular antichain if and only if it contains a periodic antichain.*

▶ **Theorem 6.** *Let $\mathcal{C}$ be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

1. *$\mathcal{C}$ is not 2-well-quasi-ordered,*
2. *There exists a class $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width which is not 2-well-quasi-ordered,*
3. *There exists a regular antichain in $\mathcal{C}$,*
4. *There exists a periodic antichain in $\mathcal{C}$,*
5. *$\mathcal{C}$ existentially transduces the class of all finite paths.*

Proof of Theorem 6. First, Item 5 implies Item 1 by Lemma 36 and the fact that 2-well-quasi-orderings and $\forall$-well-quasi-orderings coincide in our setting thanks to Theorem 4.

Then, Item 1 $\iff$ Item 2 was obtained by our analysis in Corollary 32. This analysis was refined to prove that Item 1 implies Item 3 in Lemma 35. Then, we proved in Lemma 37 that one can existentially transduce paths from a regular antichain, and therefore, we obtain Item 3 implies Item 5.

Finally, the equivalence of Item 3 and Item 4 was obtained in Corollary 40. ◀

7 Concluding Remarks

We have proven that for hereditary classes of graphs of bounded clique-width, being $\forall$-well-quasi-ordered is a very robust notion, that enjoys many structural characterisations, and where most conjectures from the literature on labelled well-quasi-orderings hold true. We consider these results as a first step towards understanding the landscape of labelled well-quasi-orderings of graphs and other relational structures. Note that we proved a very strong dichotomy result for hereditary classes of graphs of bounded clique-width: either they are not 2-well-quasi-ordered or the induced subgraph relation on them is simpler than the gap-embedding relation on trees. We conjecture that this dichotomy extends to all hereditary classes of graphs: there is nothing beyond (nested) trees. We strongly believe that our techniques can be adapted to the case of finite relational structures over a finite relational signature. Another interesting direction would be to consider non-hereditary classes of graphs of bounded clique-width. In this case, we also believe that our techniques can be adapted.

A fundamental limitation of our work is that we assume classes to be of bounded clique-width. We believe that classes of graphs that are 2-well-quasi-ordered are necessarily of bounded clique-width (see Conjecture 2), and a first step would be to show that every such class is *monadically dependent*, a notion originating from model theory [5] that has been recently investigated in finite model theory with the belief that it characterises tractable model checking for first-order formulas (see for instance [16] and [17]). We actually conjecture a stronger statement, that does not involve labelling the graphs of the class.

▶ **Conjecture 41.** *Every hereditary class of graphs that is well-quasi-ordered by the induced subgraph relation is* monadically dependent*.*

Even in the case of classes having bounded clique-width, there are several unanswered conjectures that are refinement of our results. For instance, can we existentially interpret all paths in every hereditary class of graphs that is not $\forall$-well-quasi-ordered (and of bounded clique-width). An argument similar to Lemma 36 shows that if a class existentially interprets all paths, then it is not 2-well-quasi-ordered, but the converse direction remains open.

Let us finish by discussing refinements of the notions of labelled graphs that could be considered in future works. We said that a class $\mathcal{C}$ of graphs is 2-well-quasi-ordered whenever $\mathsf{Label}_X(\mathcal{C})$ is well-quasi-ordered by the induced subgraph relation, where X is a set of two incomparable labels. There is a whole independent hierarchy of variants of k-well-quasi-ordering that are defined by using not k incomparable labels, but a set of k labels with some order on them. It is *a priori* weaker to say that $\mathsf{Label}_{a\leq b}(\mathcal{C})$ is well-quasi-ordered than to say that $\mathsf{Label}_{\{a,b\}}(\mathcal{C})$ is well-quasi-ordered. One can further restrict the labelling by asking that some labels are only used *once* in the graphs, i.e., are acting as constants distinguishing some vertices. It is known that a class $\mathcal{C}$ that is 2-well-quasi-ordered is well-quasi-ordered when finitely many constants are added to distinguish nodes [6, Lemma 5.2], and that it is still well-quasi-ordered when freely labelling the class with two comparable labels $a \leq b$. We conjecture that the converse holds, i.e., that these potential alternatives to 2-well-quasi-ordering all collapse to it.

▶ **Conjecture 42.** *Every hereditary class of graphs that is well-quasi-ordered by the induced subgraph relation with* two *additional constants is 2-well-quasi-ordered.*

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